CHAPTER 11 · OUTPUT AND COSTS · WORKED EXAMPLE

Qamar Bakery, *step by step*

One kitchen, a team that grows one baker at a time, and a daily wage bill. Follow Qamar Bakery from its first baker to its fifth, measure what each baker adds, and then turn those cakes into the bakery's short-run cost curves.



Qamar Bakery
Cakes, baked fresh every day

KITCHEN AND OVENS	EACH BAKER
960 SAR per day, fixed	360 SAR per day

OUTPUT WITH 1 TO 5 BAKERS
12 · 30 · 40 · 48 · 54 cakes per day

1 The setup

Qamar Bakery makes cakes in one rented kitchen. The kitchen, its ovens and its equipment cannot be changed quickly, but the owner decides each day how many bakers to employ. To keep the example simple, labor is the only input the bakery can vary.

The plant: fixed

Rent and equipment cost **960 SAR a day**. The bakery pays this amount whatever it bakes, even on a day with no cakes at all.

Labor: variable

Each baker earns a wage of **360 SAR a day**. The owner can employ anywhere from 0 to 5 bakers.

Output

Output is measured in **cakes per day**. How many cakes the bakery makes depends on how many bakers work in the kitchen.

Two decision time frames

The short run

The **short run** is a time frame in which the quantity of at least one resource used in production is fixed.

For Qamar Bakery, the kitchen with its ovens is fixed, and only the number of bakers can change. Such decisions are easy to reverse: a baker added for the Eid season can stop working at the bakery afterward.

The long run

The **long run** is a time frame in which the quantities of all resources can be varied, including the size of the plant.

The bakery could move to a larger kitchen or install more ovens. Decisions like these are hard to reverse. This page stays in the short run.

Quick question. Last year the bakery paid 25,000 SAR for a decorating counter built to fit its kitchen. The counter cannot be resold. Should those 25,000 SAR affect how many bakers it employs today?

Yes: the bakery spent a lot on the counter, so it should use it more.

No: that money is gone whatever the bakery decides today.

Sunk cost

A **sunk cost** is a cost that the firm has already paid and cannot recover. Because no decision made today can bring it back, a sunk cost is **irrelevant** to the firm's current decisions.

2 Hire the next baker

The kitchen is fixed, so in the short run the only way to make more cakes is to employ more bakers. Three measures connect output with the quantity of labor, L:

Total product (TP)

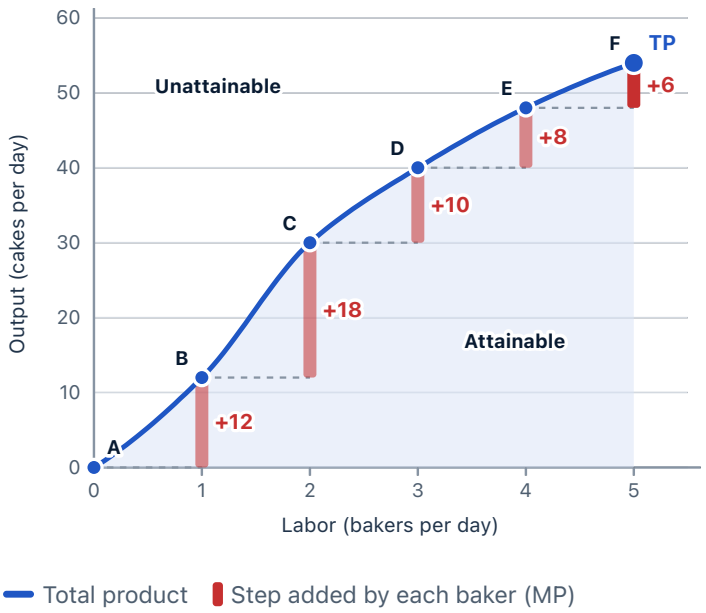
The maximum quantity that a given number of workers can produce in a given period: here, cakes per day.

Marginal product (MP)

The change in total product when one more worker is employed and all other inputs stay the same: $MP = \Delta TP \div \Delta L$.

Average product (AP)

Total product divided by the number of workers: $AP = TP \div L$.



Point	Bakers (*L*)	Total product (cakes)	Average product (cakes per baker)	Marginal product (cakes per extra baker)
A	0	0		
				12
B	1	12	12	
				18
C	2	30	15	
				10
D	3	40	13.3	
				8
E	4	48	12	
				6
F	5	54	10.8	

Each marginal value sits between two rows, because it measures the change from one row to the next. For example, 18 sits between the 1-baker and 2-baker rows: it is the 2nd baker's marginal product.

DIMINISHING MARGINAL RETURNS

5th baker: +6 cakes (from 4 to 5 bakers). Total product still rises, to 54 cakes, but each new baker adds less than the one before. Points on or below the curve are attainable with this kitchen; points above it are not.

The law of diminishing returns

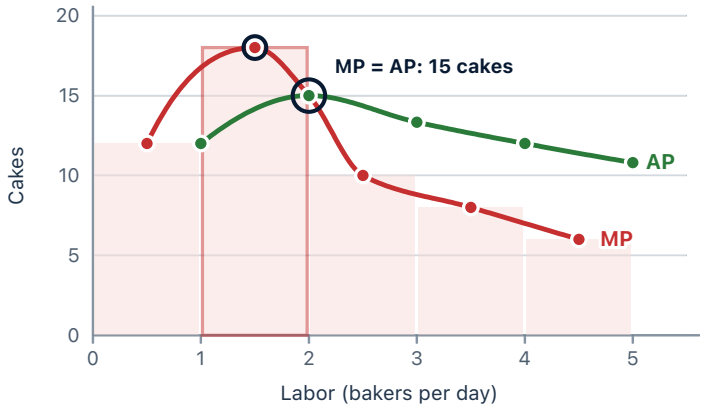
As a firm uses more of a variable input with a given quantity of fixed inputs, the marginal product of the variable input eventually diminishes.

At Qamar Bakery, the variable input is labor and the fixed input is the kitchen. Diminishing returns set in with the 3rd baker.

3 Marginal product and average product

Take the five steps from Section 2 and stand them side by side on the horizontal axis. Each bar is one baker wide and as tall as that baker's marginal product. Then compare marginal product with average product, the number of cakes per baker.

MP crosses AP at the peak of AP. With 2 bakers, marginal product equals average product, 15 cakes, and average product is at its highest. To the left MP is above AP; to the right it is below.



Marginal product (MP) Average product (AP)

Hover over or tap the graph, or use the arrow keys, to read each baker's values.

Point	Bakers (*L*)	Total product (cakes)	Average product: *TP* ÷ *L*	Margir produ change i
A	0	0		
B	1	12	12 ÷ 1 = **12**	12 – 0 =
C	2	30	30 ÷ 2 = **15**	30 – 12
D	3	40	40 ÷ 3 = **13.3**	40 – 30
E	4	48	48 ÷ 4 = **12**	48 – 40
F	5	54	54 ÷ 5 = **10.8**	54 – 48

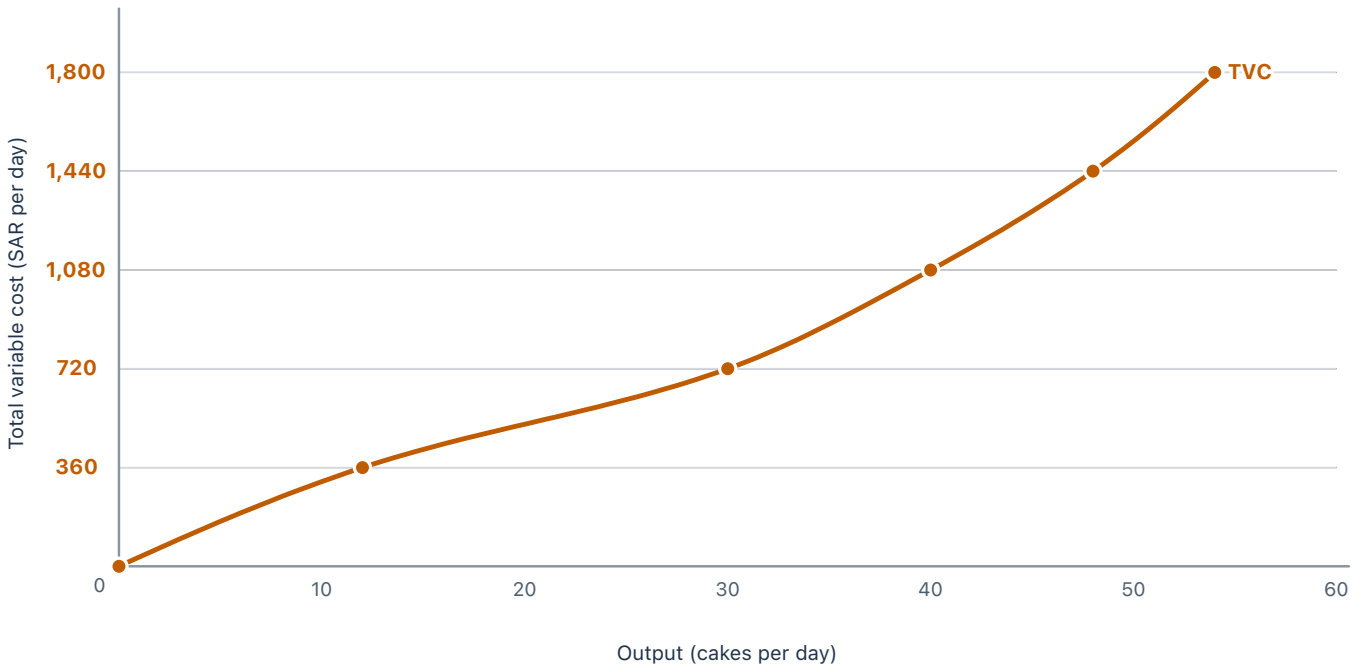
Each marginal value sits between two rows, because it measures the change from one row to the next. Highlighted cells are the points just plotted.

2nd baker: +18 cakes (from 1 to 2 bakers). Average product with 2 bakers: **15** cakes per baker. The 2nd baker adds more than the previous average of 12, so AP rises.

The marginal product of each new baker pulls the average toward itself, in the same way that one new mark changes a class average. A baker who adds more than the current average raises it; a baker who adds less lowers it.

4 Pay the bakers

Each baker costs 360 SAR a day, so every number of bakers comes with a total variable cost: ***TVC* = 360 × *L***. One baker costs 360 SAR, two cost 720 SAR, and so on. Write those costs under the labor axis, then swap the axes so that output is on the horizontal axis and cost is on the vertical axis.



The axes are swapped: output is on the horizontal axis and total variable cost on the vertical axis. The total product curve has become the **TVC curve**. For example, 30 cakes need 2 bakers, so they cost 720 SAR in wages.

- The **TVC curve** is the total product curve flipped over: the same pairs of numbers, with the axes exchanged.
- Where the TP curve gets **steeper** (increasing marginal returns), the TVC curve gets **flatter**. Where the TP curve gets **flatter** (diminishing marginal returns), the TVC curve gets **steeper**.

5 Total costs

With the plant fixed, the bakery can bake more only by paying for more bakers, so its costs climb as output grows. Economists use three measures of total cost.

Total fixed cost (TFC)

The cost of the firm's fixed inputs. It does **not** change with output: 960 SAR a day for the bakery.

Total variable cost (TVC)

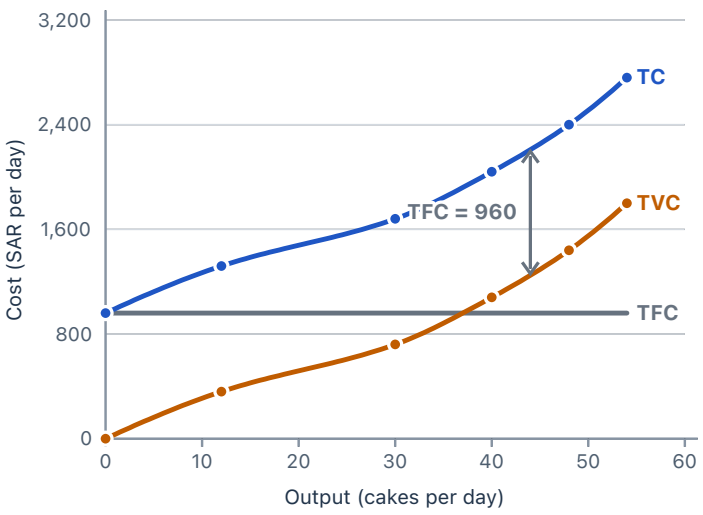
The cost of the firm's variable inputs. It **does** change with output: 360 SAR for each baker.

Total cost (TC)

The cost of all the resources the firm uses: $TC = TFC + TVC$.

Total fixed cost (rent and equipment) **960 SAR**

960 1,040 1,120 1,200



Bakers	Output (cakes)	TFC	TVC	TC
0	0	960	0	960
1	12	960	360	1,320
2	30	960	720	1,680
3	40	960	1,080	2,040
4	48	960	1,440	2,400
5	54	960	1,800	2,760

All costs in SAR per day.

TFC TVC TC

Move the slider to raise the rent. Watch which curves move and which stay put.

- The **TFC curve** is horizontal, because fixed cost is the same at every output.
- The **TC curve** is the TVC curve shifted up by total fixed cost. The vertical gap between them is always TFC.
- TVC and TC rise **slowly at first and steeply later**: from 12 to 30 cakes, each extra cake adds 20 SAR to cost instead of 30, because the 2nd baker makes the team more productive; beyond 30 cakes, diminishing returns make each extra cake more expensive.

6 Marginal and average costs

Marginal cost (MC)

The increase in total cost that results from a one-unit increase in output: $MC = \Delta TC \div \Delta Q$.

With labor as the only variable input, MC = wage $\div$ MP.

Average fixed cost (AFC)

Total fixed cost per unit of output: $AFC = TFC \div Q$.

Average variable cost (AVC)

Total variable cost per unit of output: $AVC = TVC \div Q$.

Average total cost (ATC)

Total cost per unit of output: *ATC* = *TC* ÷ *Q*. Because *TC* = *TFC* + *TVC*, *ATC* = *AFC* + *AVC*.

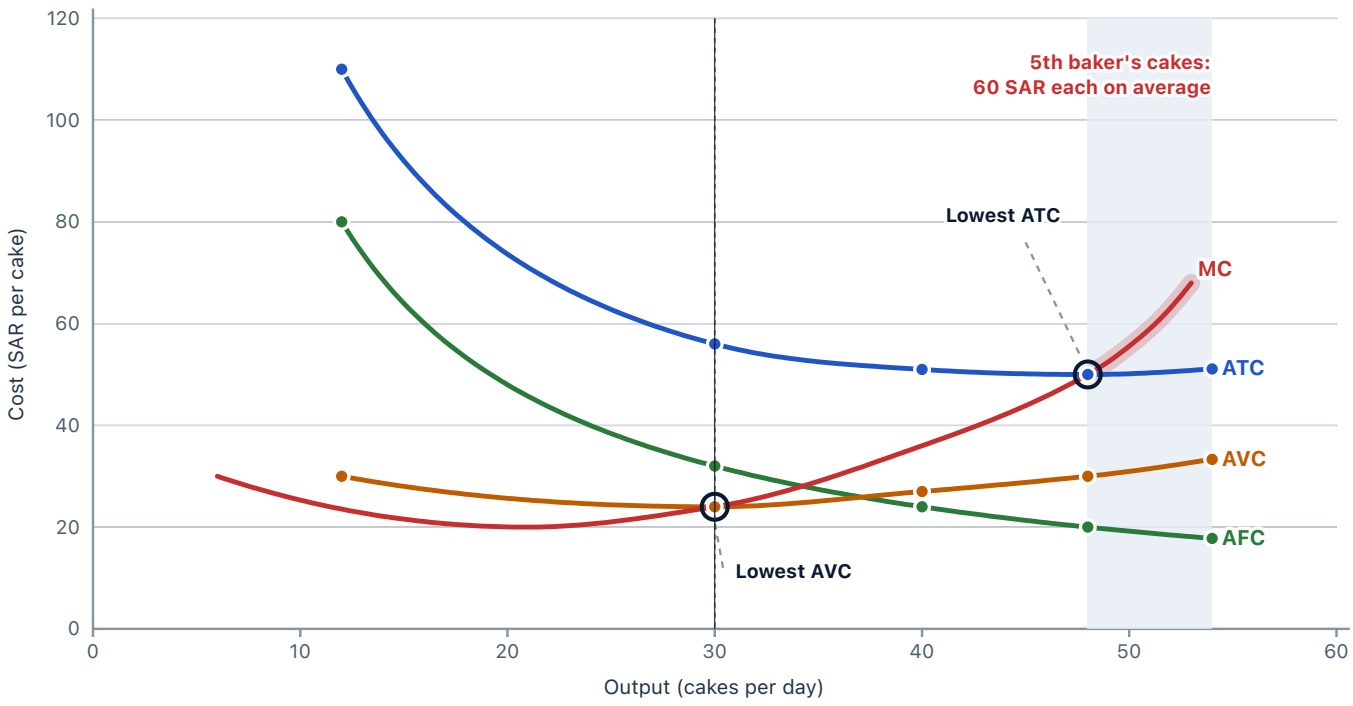
Bakers	Output, *Q* (cakes)	TC = TFC + TVC	AFC	AVC	ATC	MC (per extra cake)
0	0	960 + 0 = **960**				
						360 ÷ 12 = **30**
1	12	960 + 360 = **1,320**	960 ÷ 12 = **80**	360 ÷ 12 = **30**	1,320 ÷ 12 = **110**	
						360 ÷ 18 = **20**
2	30	960 + 720 = **1,680**	960 ÷ 30 = **32**	720 ÷ 30 = **24**	1,680 ÷ 30 = **56**	
						360 ÷ 10 = **36**
3	40	960 + 1,080 = **2,040**	960 ÷ 40 = **24**	1,080 ÷ 40 = **27**	2,040 ÷ 40 = **51**	
						360 ÷ 8 = **45**
4	48	960 + 1,440 = **2,400**	960 ÷ 48 = **20**	1,440 ÷ 48 = **30**	2,400 ÷ 48 = **50**	
						360 ÷ 6 = **60**
5	54	960 + 1,800 = **2,760**	960 ÷ 54 = **17.8**	1,800 ÷ 54 = **33.3**	2,760 ÷ 54 = **51.1**	

Costs in SAR. Each marginal value sits between two rows, because it measures the change from one row to the next. MC = 20 sits between the 12-cake and 30-cake rows: it is the cost of each extra cake made by the 2nd baker, averaged over that baker's cakes. The MC curve is a smooth illustration of how marginal cost changes cake by cake.

Row 5. The 5th baker costs 360 SAR and adds 6 cakes (from 48 to 54 cakes), so the 5th baker's cakes cost **360 ÷ 6 = 60 SAR** each. At 54 cakes: AFC = 960 ÷ 54 = 17.8, AVC = 1,800 ÷ 54 = 33.3, ATC = 2,760 ÷ 54 = 51.1 SAR.

On the graph: new dots for AFC, AVC and ATC at 54 cakes, the average cost curves extended to 54 cakes, and the MC curve over the 5th baker's cakes, from 48 to 54, highlighted.

All rows are in. Check that ATC = AFC + AVC in every row: for example, 32 + 24 = 56.



MC AFC AVC ATC Hover over or tap the graph, or use the arrow keys, to read the values.

30 cakes (2 bakers): AFC **32**, AVC **24**, ATC **56** SAR. The MC curve here is at about 24 SAR. These cakes are the 2nd baker's cakes, from 12 to 30; in the table they cost 20 SAR each on average. AVC is at its minimum, and MC passes through it.

Marginal cost and the averages

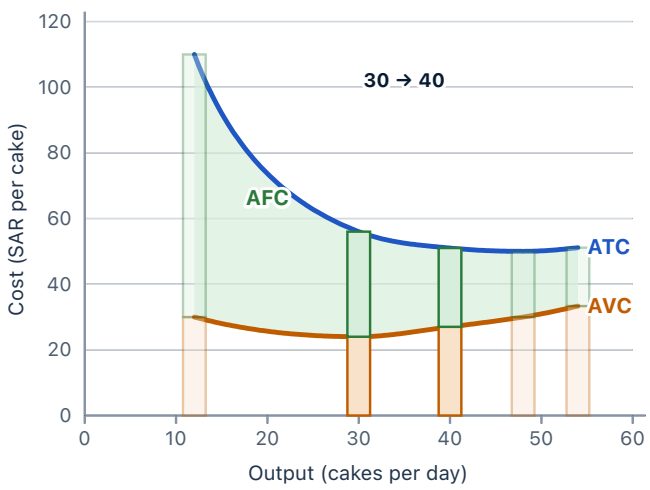
Where **MC is below** an average cost, it **pulls that average down**. Where **MC is above** it, it **pulls the average up**. So MC passes through the lowest point of the AVC curve (24 SAR at 30 cakes) and the lowest point of the ATC curve (50 SAR at 48 cakes).

Example: the 2nd baker's cakes cost 20 SAR each, less than the AVC of 30 SAR at 12 cakes, so AVC falls to 24 SAR. The 3rd baker's cakes cost 36 SAR each, more than 24 SAR, so AVC rises to 27 SAR.

7 Why the average cost curves are U-shaped

AVC moves opposite to average product, because *AVC* = wage ÷ *AP*. While AP rises, AVC falls; with 2 bakers each baker makes 15 cakes on average, so the wage cost per cake is 360 ÷ 15 = 24 SAR. Once AP falls, AVC rises: with 4 bakers, AVC = 360 ÷ 12 = 30 SAR.

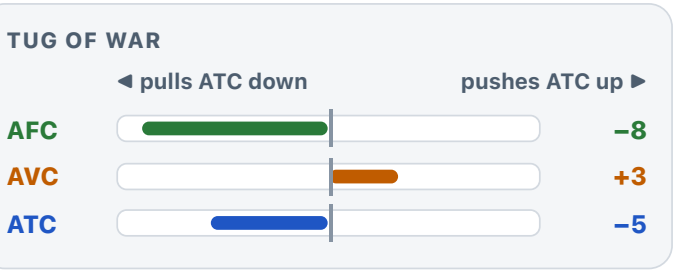
ATC is AFC stacked on top of AVC. Two forces pull it in opposite directions: spreading the 960 SAR of fixed cost over more cakes pushes AFC down, while diminishing returns eventually push AVC up. Pick a step and watch the tug of war.



AVC part AFC part ATC = AFC + AVC

Output (cakes)	AFC	AVC	ATC = AFC + AVC
30	32	24	32 + 24 = **56**
40	24	27	24 + 27 = **51**
48	20	30	20 + 30 = **50**
54	17.8	33.3	17.8 + 33.3 = **51.1**

From 30 to 40 cakes: AFC 32 → 24 (−8), AVC 24 → 27 (+3), so ATC 56 → 51 (−5): **the fall in AFC wins**, and ATC falls.



Change in output	Change in AFC	Change in AVC	Change in ATC
30 to 40 cakes	−8	+3	**−5**
40 to 48 cakes	−4	+3	**−1**
48 to 54 cakes	−2.2	+3.3	**+1.1**

Changes in SAR per cake.

- Up to 48 cakes, the fall in AFC outweighs the rise in AVC, so ATC falls.
- At high output, AFC hardly falls any more, while diminishing returns keep pushing AVC up. Beyond 48 cakes the rise in AVC wins, and ATC rises.

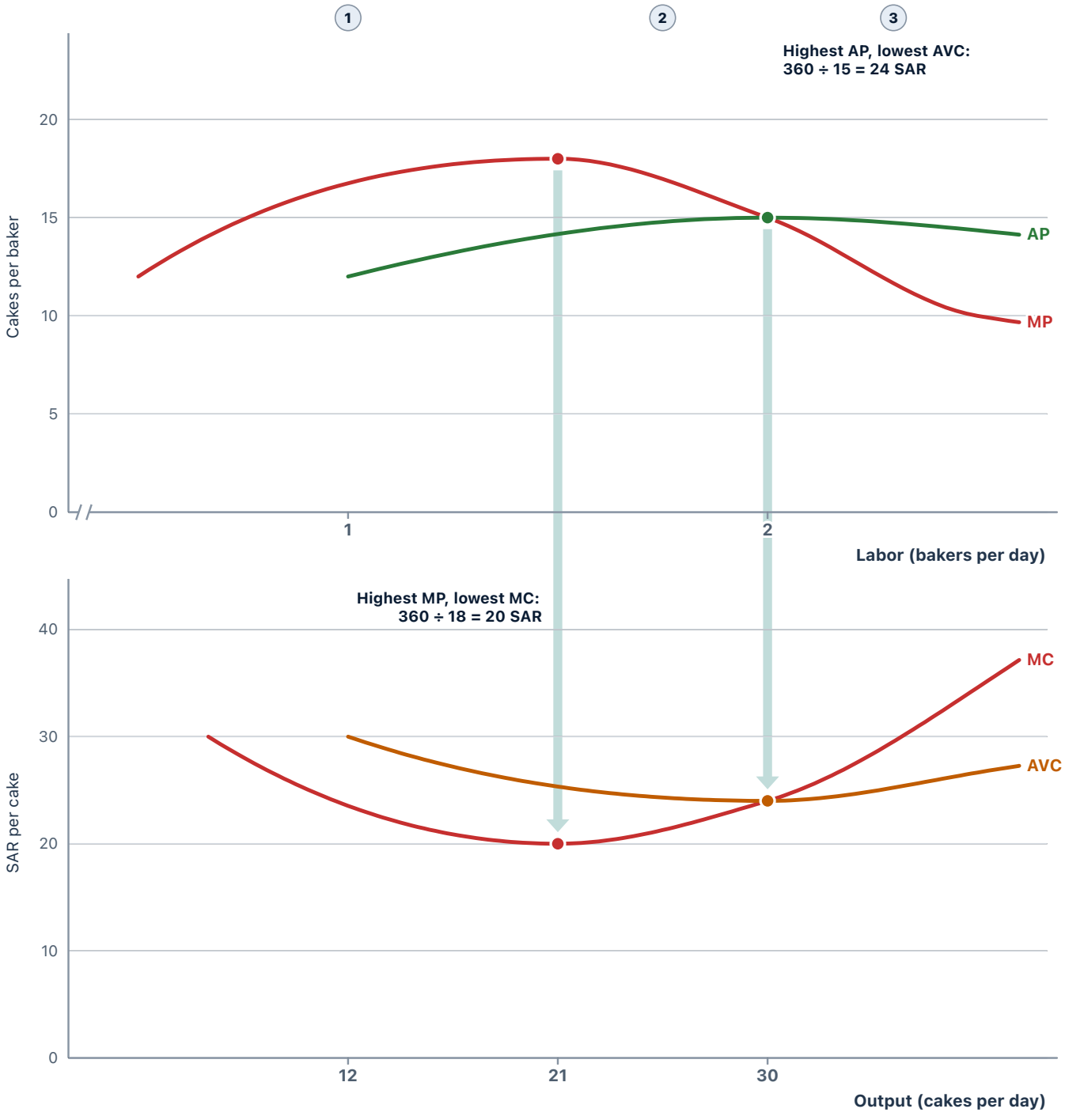
8 The mirror: product curves and cost curves

The bakery's technology, summarized by its product curves, decides the shape of its cost curves. The link is the wage: ***MC* = wage ÷ *MP*** and ***AVC* = wage ÷ *AP***. So when a product curve is high, the matching cost curve is low.

Here the product graph sits directly above the cost graph. The horizontal scales are set so

that 1 baker is directly above 12 cakes and 2 bakers directly above 30 cakes, the output each team makes. Both graphs stop at about 41 cakes: further right the two scales no longer match, because 3 bakers would sit above 48 cakes but make only 40. Sections 3 and 6 show the full curves.

1. MP rises, so MC falls. AP rises, so AVC falls.
2. MP falls, so MC rises. AP still rises, so AVC still falls.
3. MP and AP both fall, so MC and AVC both rise.



MP and MC AP AVC Links between matching points

Choose a link or a region above, or move over the graphs. The pale arrows join each product peak to the matching cost minimum, and each number marks a region of both graphs.

- **MP and MC.** Marginal cost is lowest where marginal product is highest. Over the range where MP rises, MC falls; where MP falls, MC rises.
- **AP and AVC.** AVC is lowest where average product is highest: with 2 bakers, AP peaks at 15 cakes per baker, and AVC reaches its minimum of 24 SAR at the matching output of 30 cakes.

9 Shift sandbox

Two things decide where the cost curves sit: the firm's **technology** and the **prices of the factors of production** it buys. Change the wage, the fixed cost or the bakers' productivity and see which curves move. The dashed curves show the base case.

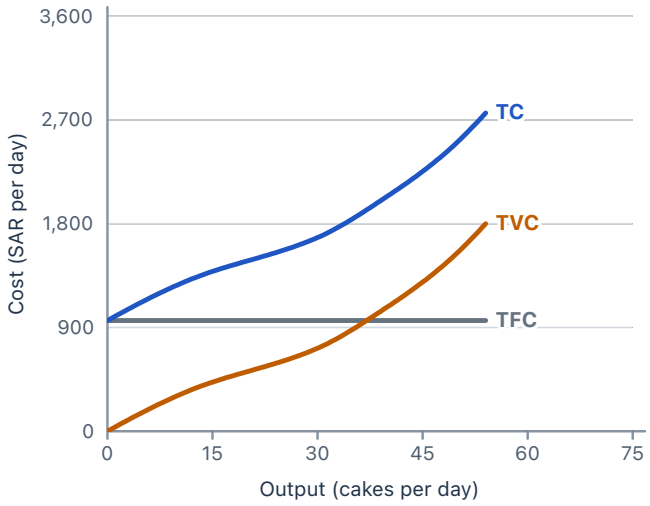
Wage per baker 360 SAR
300 360 450

Total fixed cost 960 SAR
960 1,080 1,200

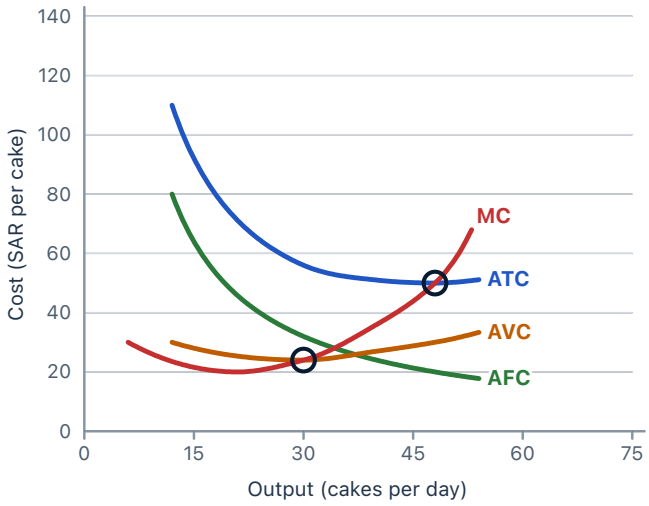
Extra cakes per baker +0%
0% 10% 20% 30%

TFC	TVC	TC	AFC	AVC	ATC
no change	no change	no change	no change	no change	no change

MC
no change



TFC TVC TC



MC AFC AVC ATC

Lowest AVC: **24 SAR at 30 cakes**, where MC crosses it (circled). Lowest ATC: **50 SAR at 48 cakes**, where MC crosses it (circled). This is the base case. Move a slider or pick a scenario.

Bakers	Output (cakes)	TC	AFC	AVC	ATC	MC
0	0	960				
						30
1	12	1,320	80	30	110	
						20
2	30	1,680	32	24	56	
						36
3	40	2,040	24	27	51	
						45
4	48	2,400	20	30	50	
						60
5	54	2,760	**17.8**	**33.3**	**51.1**	

Costs in SAR; values rounded to one decimal place. Changed values are highlighted. Each marginal value sits between two rows, because it measures the change from one row to the next.

- **Higher productivity** moves the product curves up and the cost curves down: every cake needs less labor.
- **A higher fixed cost** shifts TFC, TC, AFC and ATC up, but leaves TVC, AVC and MC where they were. The cost of each extra cake does not change.
- **A higher wage** shifts TVC, TC, AVC, ATC and MC up, and leaves TFC and AFC unchanged.

10 Check yourself

Choose an answer for each question. You get feedback at once.

1. With 3 bakers the bakery makes 40 cakes a day; with 4 bakers it makes 48 cakes. What is the 4th baker's marginal product?

8 cakes | 12 cakes | 48 cakes | 10 cakes

Answer: 8 cakes. Marginal product is the change in total product: 48 – 40 = 8 cakes. (48 cakes is total product with 4 bakers, and 48 ÷ 4 = 12 cakes is their average product.)

2. Each baker earns 360 SAR a day. With 1, 2, 3 and 4 bakers the bakery makes 12, 30, 40 and 48

2. Each baker earns 360 SAR a day. With 1, 2, 3 and 4 bakers the bakery makes 12, 30, 40 and 48 cakes. At which of these outputs is average variable cost (TVC ÷ Q) lowest?

- 12 cakes
- 30 cakes
- 40 cakes
- 48 cakes

Answer: 30 cakes. AVC = TVC ÷ Q: 360 ÷ 12 = 30, 720 ÷ 30 = 24, 1,080 ÷ 40 = 27 and 1,440 ÷ 48 = 30 SAR. It is lowest, 24 SAR, at 30 cakes, where average product is highest (30 ÷ 2 = 15 cakes per baker, and 360 ÷ 15 = 24).

3. The rent rises, and total fixed cost goes from 960 to 1,200 SAR a day. Which curves shift?

- TFC, TC, AFC and ATC; MC and AVC stay put
- Every cost curve, including MC
- Only MC and AVC
- None, because rent is a sunk cost

Answer: TFC, TC, AFC and ATC; MC and AVC stay put. Rent is a fixed cost, so TFC and TC shift up by 1,200 – 960 = 240 SAR, and AFC and ATC shift up too. The wage bill for each number of bakers is unchanged, so TVC, AVC and MC do not move. Rent is a fixed cost, not a sunk cost: the bakery pays it every day it keeps the kitchen.

4. Each baker earns 360 SAR a day. With 1 baker the bakery makes 12 cakes; with 2 bakers it makes 30 cakes. What is the marginal cost of each cake the 2nd baker adds?

- 30 SAR
- 24 SAR
- 20 SAR
- 12 SAR

Answer: 20 SAR. The 2nd baker adds 360 SAR to cost and 30 – 12 = 18 cakes to output, so MC = 360 ÷ 18 = 20 SAR per cake. 30 SAR is the 1st baker's cakes (360 ÷ 12), 24 SAR is AVC at 30 cakes (720 ÷ 30), and 12 SAR divides the wage by total output instead of the extra output.

5. From 48 to 54 cakes, AFC falls from 20 to 17.8 SAR and AVC rises from 30 to 33.3 SAR. Why does average total cost rise beyond 48 cakes?

- Fixed cost rises when output rises
- AFC starts to rise
- Diminishing returns push AVC up by more than AFC falls
- The wage rises for extra bakers

Answer: Diminishing returns push AVC up by more than AFC falls. AFC still falls, but only by 2.2 SAR, while diminishing returns push AVC up by 3.3 SAR. The rise in AVC wins, so ATC rises from 20 + 30 = 50 to 17.8 + 33.3 = 51.1 SAR, by 1.1 SAR.

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